

COURSE NOTES

TOPICS IN ALGEBRA: GROUP THEORY

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Chapter 1

Lecture 1

Chapter 2

Lecture 2

Recall last time:

1. Matrix Lie algebras, e.g. $\mathfrak{sl}(n)$. Consider involution $*$: $M_n(F) \rightarrow M_n(F)$ skew-symmetric matrices.

2. Derivation, $\text{Der}(A)$. e.g. $A = F[t_1, \dots, t_n]$, $d = \sum_{i=1}^n f_i(t) \partial_{t_i}$

A non-associative algebra, $d : x \mapsto [a, x]$ is a derivation.

Let L a Lie algebra, inner derivation

Q: d a derivation and $\exp(d)$. It need convergence.

Example 2.0.1. $d = t_1 t_2 \frac{\partial}{\partial t_1}$.

3. Brackets:

2.1 Engel Theorem

L^n : span of products of elements of length n . Recall $\text{ad}([a, b]) = [\text{ad}(a), \text{ad}(b)]$. $L \rightarrow \text{End}_F(L)^{(-)}, a \mapsto \text{ad}(a)$.

$[I, J]$ is again an ideal.

Decending chain:

$$L = L^1 \supseteq L^2 \supseteq \dots$$

L is nilpotent if...

eg: strict upper matrix

Another decending chain: $L^{[1]} = [L, L]$, $L^{[n+1]} = [L^{[n]}, L^{[n]}]$

L is solvable if

e.g. 2 dimentional sovable non nilpotent Lie alg.

A associa. $L \subseteq A^{(-)}$. Suppose $A = \langle L \rangle$ as an associa alge. Then A is an enveloping algebra of L .

Theorem 2.1.1. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A = \langle L \rangle$. $\exists n \geq 1, \forall a \in L, a^n = 0$. Then A is nilpotent.*

We say that a subset $S \subseteq L$ is a Lie subset if $\forall a, b \in S, [a, b] \in S$.

Theorem 2.1.2. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A = \langle L \rangle$. Let S be a Lie subset of L . $L = \text{span}(S)$, $\forall a \in S$, $a^n = 0$.*

Choose a maximal Lie subset $S_1 \subset S$ such that the associative $\langle S_1 \rangle$ is nilpotent. So $\exists n$ such that $S_1 \dots S_1 = (0)$ (n times) (existence?)
 claim $\text{ad}(S_1) \dots \text{ad}(S_n) = (0)$.